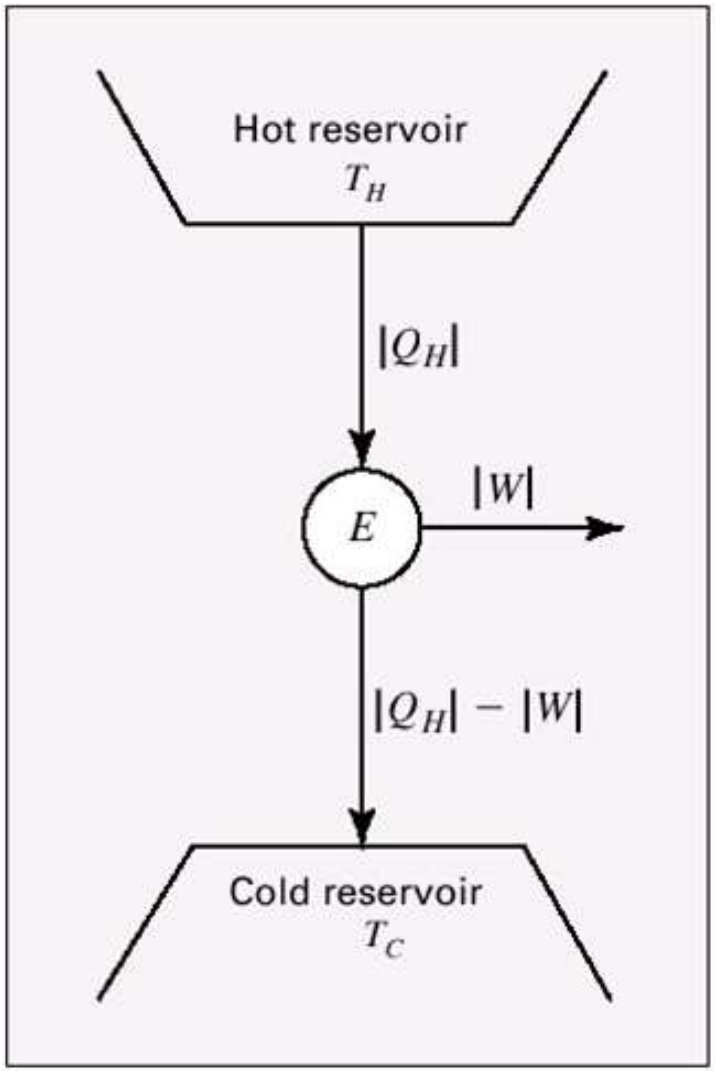
CH365 Chemical Engineering Thermodynamics

Lesson 25

3rd Law and Entropy from the Microscopic Viewpoint

What is Entropy?

Thermal Efficiency



(Eq. 5.8)

$$\eta = \frac{|W|}{|Q_H|} = 1 - \frac{T_C}{T_H}$$

Third Law of Thermodynamics

Untitled-1 * - Wolfram Mathematica 13.1

File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

Input

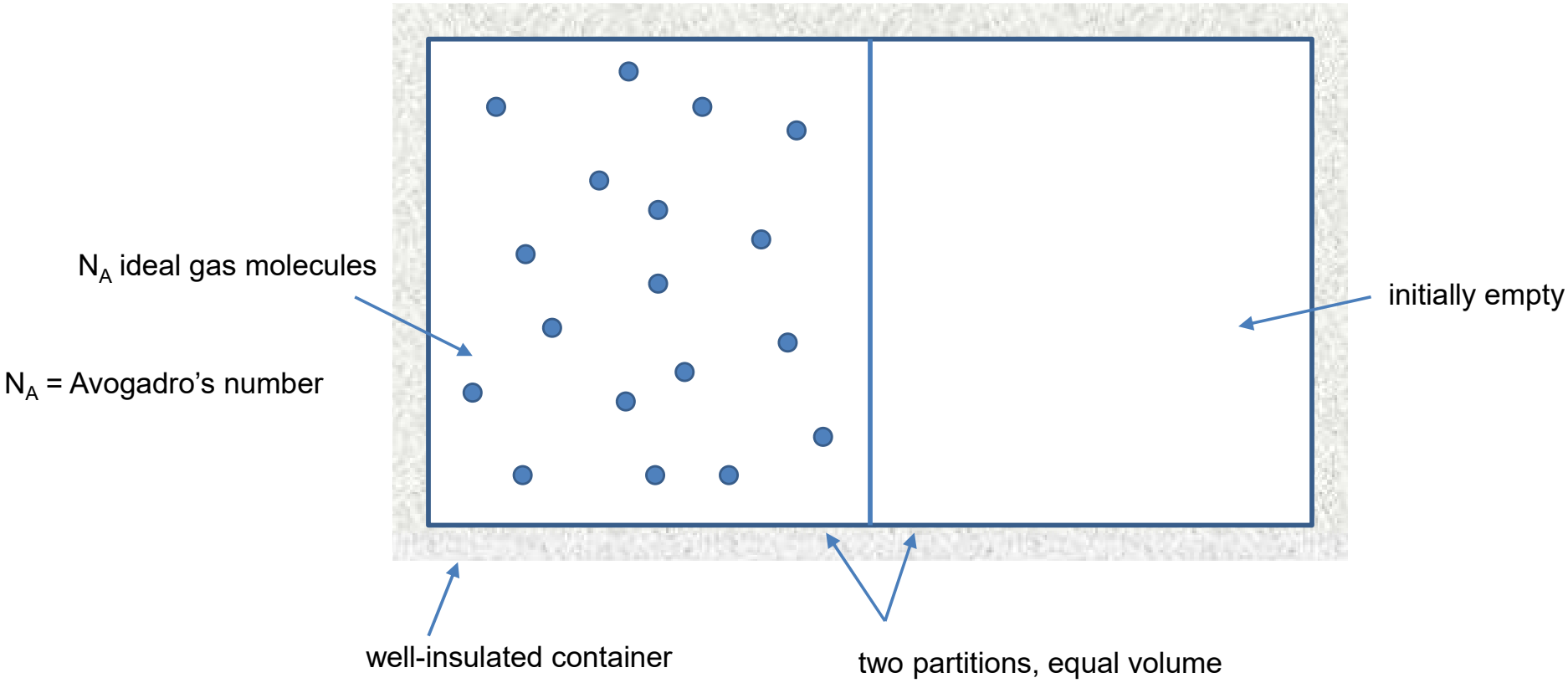
In[1]:=
```
(*CaO(s) m.p. 2845 K*)
(*Cp from Appendix C page 670*)
Cp = 6.104 + .443 * 10^-3 * T - 1.047 * 10^5 * T^-2;
```

In[2]:=
$$\int_0^{273} \frac{Cp}{T}\, dT$$

100%

Statistical Interpretation

- ideal gas
  - molecules do not interact
  - internal energy resides within the individual molecules



$$U = U(T,V)$$

ch. 4, p. 138, (un-numbered equation)

$$dU = \left(\frac{\partial U}{\partial T}\right)_V dT + \underbrace{\left(\frac{\partial U}{\partial V}\right)_T}_{0} dV$$

0 → Because molecules don't interact

definition of C_V

$$\frac{\Delta S}{R} = \underbrace{\int_{T_{before}}^{T_{after}} \frac{C_P^{ig}}{R}\frac{dT}{T}}_{0} - \underbrace{\ln\left(\frac{P_{final}}{P_{initial}}\right)}_{}$$

0, no change in T

P drops by half

(Eq. 5.10)

ch. 3, p. 79, (Eq. 3.13a)

$$dU = C_V dT$$

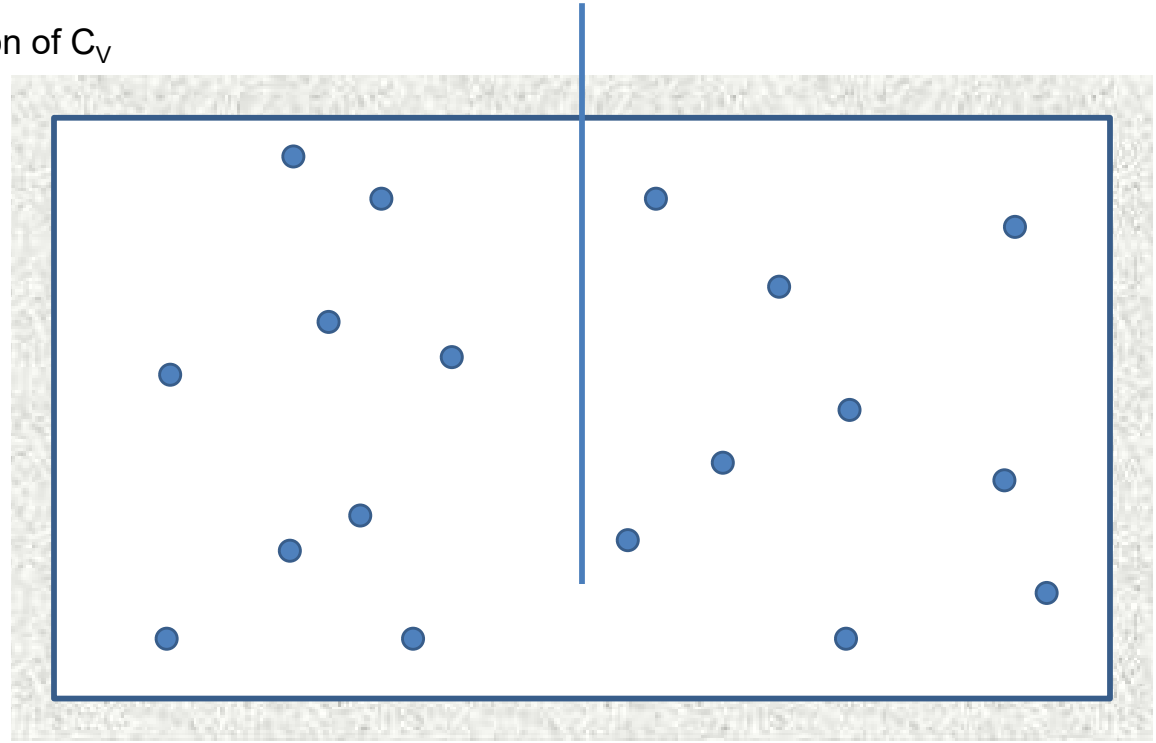
Recall:

ch. 3, p. 79, (Eq. 3.14a)

$$dH = C_P dT$$

definition of C_P:

$$C_P \equiv \left(\frac{\partial H}{\partial T}\right)_P$$



$$\Delta U = C_V \Delta T = 0$$

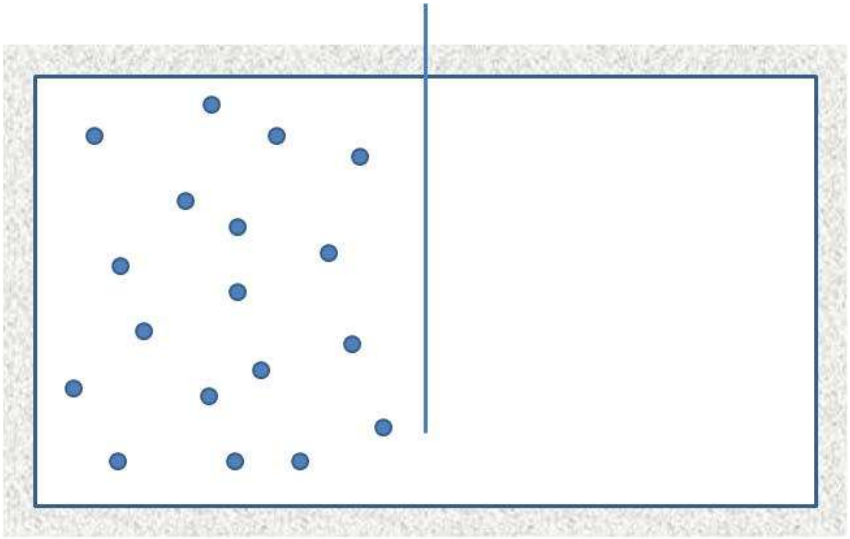
But if $\Delta U=0$, then T does not change.

$$\Delta S = -R \cdot \ln\left(\frac{P_{after}}{P_{before}}\right) = \underbrace{R \cdot \ln(2)}_{}$$

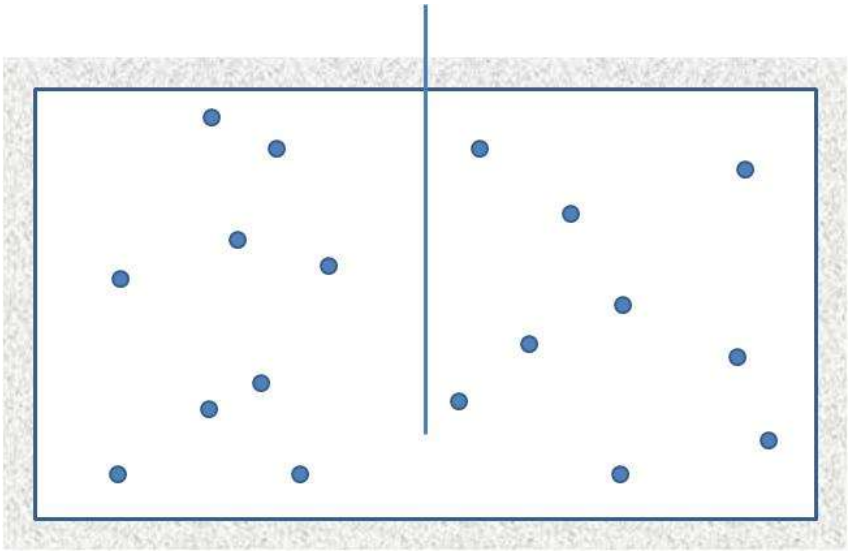
>0 (irreversible)

Result of *classical* thermodynamics

more ordered → less random → less disordered



less ordered → more random → more disordered



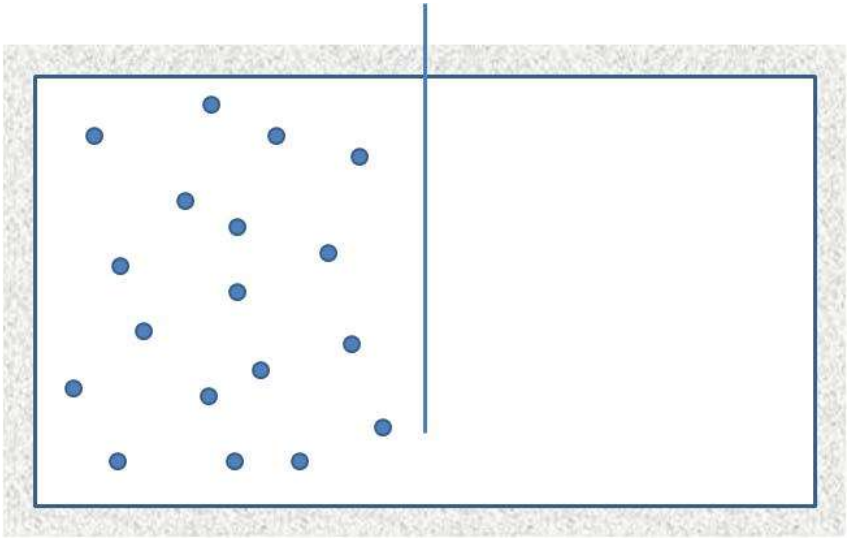
- immediately after opening
- molecules are not randomly distributed over the total volume
- crowded into half the space

Increasing disorder (or decreasing structure) on the molecular level corresponds to increasing entropy.

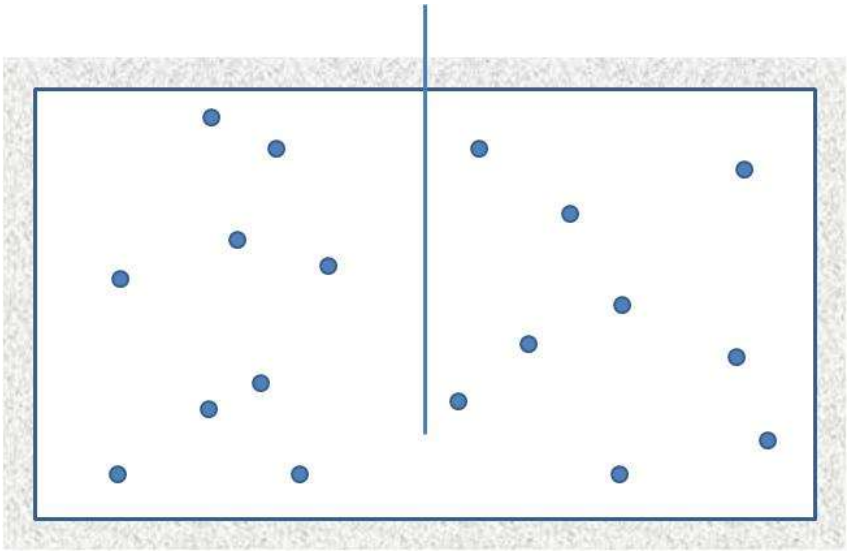
Expression for disorder postulated by J.W. Gibbs and L. Boltzmann, 1878.

Quantitative Expression of Disorder

more ordered = less random = less disordered



less ordered = more random = more disordered



All molecules are in one of the two states.

$$\Omega_{\text{initial}} = \frac{N_A!}{(N_A!)(0!)}$$

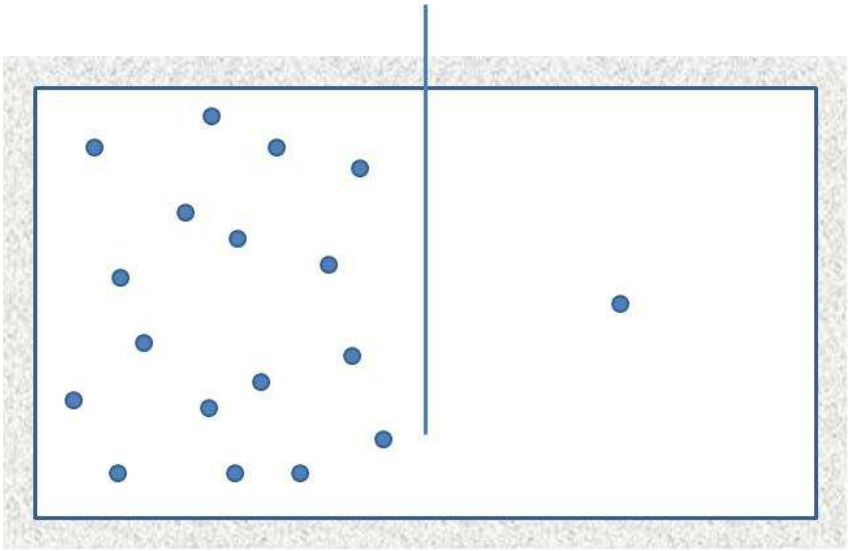
$$= \frac{18!}{(18!)(0!)}$$

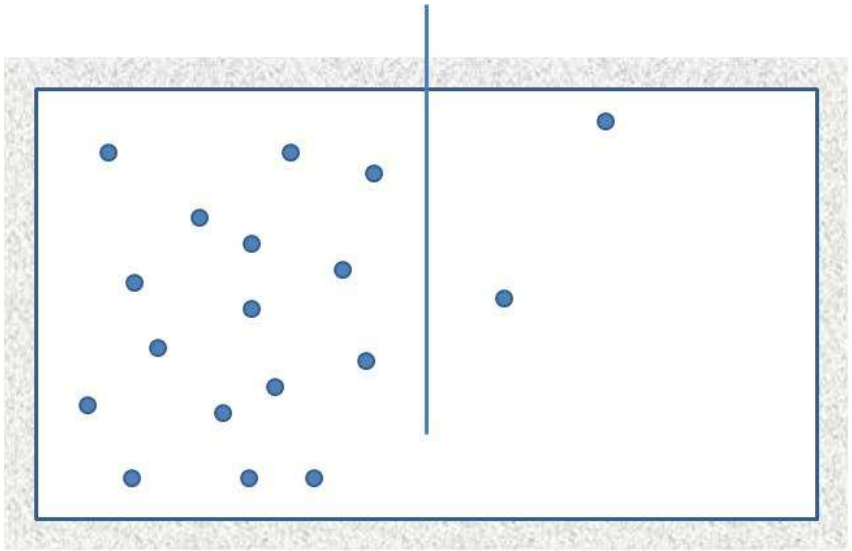
$$= 1$$

$$\Omega_{\text{final}} = \frac{N_A!}{\left(\frac{N_A}{2}!\right)\left(\frac{N_A}{2}!\right)}$$

$$= \frac{18!}{9! \cdot 9!}$$

$$= 48,620$$





$$\Omega_1 = \frac{18!}{(17!)(1!)} = 18$$

$$\Omega_2 = \frac{18!}{(16!)(2!)} = 153$$

unbounded as N_A increases

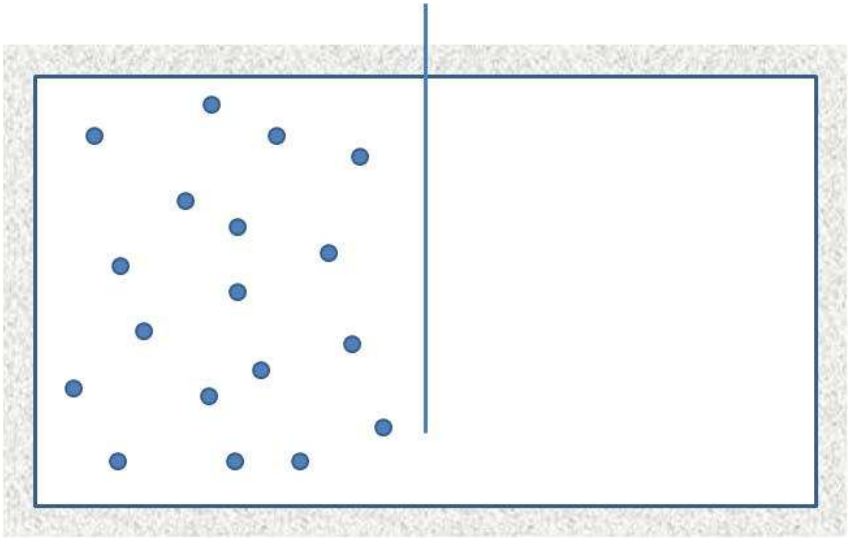
for the 18 particles, $\Omega_{final} = 48,620$

How about $N_A = 10^{23}$?

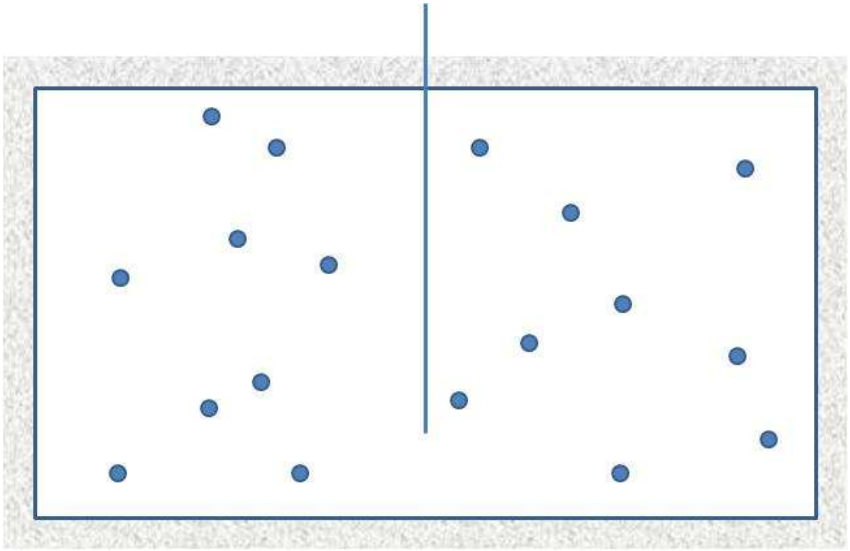
Boltzmann’s thought process:

Boltzmann’s thought process, continued:

more ordered = less random = less disordered



less ordered = more random = more disordered



- immediately after opening
- molecules are not randomly distributed over the total volume
- crowded into half the space

Increasing disorder (or decreasing structure) on the molecular level corresponds to increasing entropy.



https://en.wikipedia.org/wiki/File:Zentralfriedhof_Vienna_-_Boltzmann.JPG

Questions?